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Unit -5

Sleep and Rest Pattern

SLEEP AND REST

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- Rest and sleep are fundamental components of wellbeing
- One- third of human life is spent sleeping.
- Periods of rest may account for another major portion of the life span
- The need for rest and sleep varies with age, developmental level, health status, activity level, and cultural norms

Sleep and rest...

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- **Sleep** refers to a natural state of altered consciousness during w/c an individual experiences minimal physical activity and a general slowing of the body's physiological processes as well as decreased awareness & responsiveness to stimuli
- Sleep generally occurs in a periodic cycle and usually lasts for several hours at a time
- Disruptions in the usual sleep routine can be distressing to clients and will most likely impair sleep further

SLEEP

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- Sleep is a state of unconsciousness from which the person can be aroused by sensory stimuli
- A normal adult person sleeps 7-8 hrs/day
- A newly-born infant sleeps much longer time (16-18 hrs/day), while an old person sleeps less (5-6 hrs/day)
- It is to be distinguished from *coma*, which is unconsciousness from which the person cannot be aroused
- There are multiple stages of sleep, from very light sleep to very deep sleep

WAKING/SLEEPING RHYTHM

- Waking and sleeping periods follow each other in a circadian rhythm (i.e. 24 hr. rhythm) which is synchronized with the daily light-dark cycle
- This synchronization is the function of the suprachiasmatic nucleus of the hypothalamus which receives collateral from the visual pathway
- Physical and psychological factors affect the onset and duration of sleep.
 - E.g. cold and fear prevent the onset of sleep, whilst fatigue and bored facilitate its onset

REST

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- **Rest** refers to a state of relaxation and calmness, both mental and physical
- In rest awareness of the environment is maintained but motor or cognitive response is decreased and may involve the total body system or only a part
- Activity during rest periods can range from lying down to reading a book to taking a quiet walk
- When discussing a client's rest patterns, the nurse should try to understand what activities and environments the client defines as restful

Normal Sleep/ Rest Function

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□ Sleep and rest have the following functions:-

1. Restorative, (having the ability to restore health, strength or feeling of well-being)
2. Protective
3. Conservation of metabolic energy
4. Neural maturation,
5. Facilitation of learning or memory,
6. Cognition, (Mental action or process of acquiring knowledge through thought, experience, & senses)
7. *The principal value of sleep is to restore natural balances among the neuronal centers (E.g.. Suicide, depression)*

Normal Physiology of Sleep

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- Sleep is an active process involving the reticular activating system (RAS) and a dynamic interaction of neurotransmitters
- The RAS consists of a network of interconnecting neurons in the medulla, pons, and midbrain with projections to the spinal cord, hypothalamus, cerebellum, and cerebral cortex
- Serotonin is a major neurotransmitter associated with sleep
- It is produced in the *Raphe nuclei* of the brain stem

- Serotonin is thought to decrease the activity of the RAS, there by inducing and sustaining sleep
- Sleep-wake patterns appear to be affected by and to affect certain hormone levels
- Melatonin from the pineal gland is secreted in enormous quantities during sleep
- Secretion of growth hormone, Adreno-corticotropic hormone and prolactin increases early in the sleep period

Neuronal Centers, Neuro-humoral Substances, & Mechanisms that Cause Sleep:

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□ A Possible Specific Role for Serotonin

1. The most conspicuous stimulation area for causing almost natural sleep is the *raphe nuclei in the lower half of the pons and in the medulla*
- These nuclei comprise a thin sheet of special neurons located in the *midline*
 - Nerve fibers from these nuclei spread locally in the *brain stem* reticular formation and also upward into the *thalamus, hypothalamus*, most areas of the limbic system, and even the *neocortex* of the *cerebrum*

- In addition, fibers extend downward into the spinal cord, terminating in the posterior horns, where they can inhibit incoming sensory signals, including pain
- Many nerve endings of fibers from these **raphe neurons** secrete **serotonin**
- When a drug that blocks the formation of serotonin is administered to an animal, the animal often cannot sleep for the next several days
- Therefore, it has been assumed that serotonin is a transmitter substance associated with production of sleep

Cont...

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2. Stimulation of some areas in the *nucleus of the tractus solitarius* can also cause *sleep*
 - This nucleus is the termination in the *medulla* and *pons* for visceral sensory signals entering by way of the *vagus* and *glossopharyngeal nerves*
3. Sleep can be promoted by stimulation of several regions in the *diencephalon*, including (1) the rostral (front end) part of the *hypothalamus*, mainly in the *suprachiasmatic area*, and (2) an occasional area in the *diffuse nuclei* of the *thalamus*

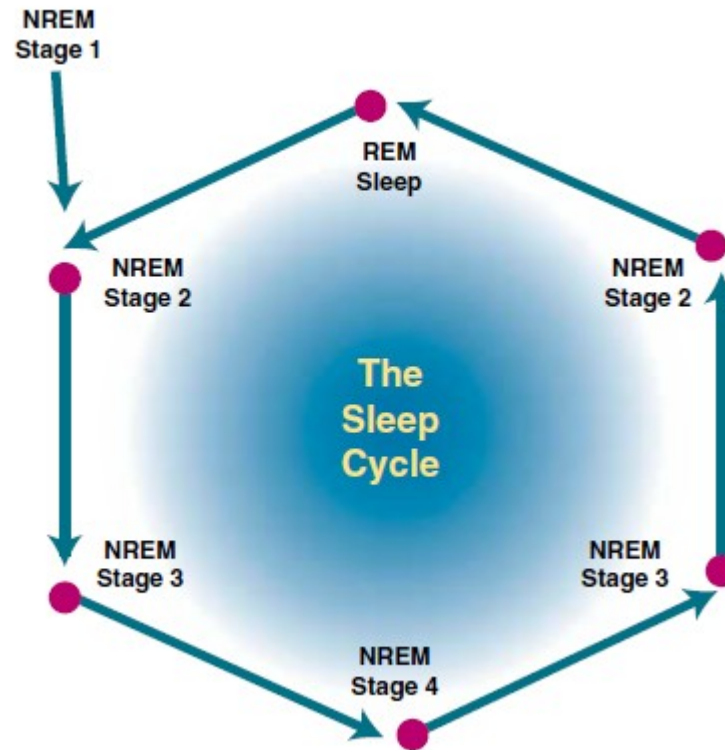
Lesions in Sleep-Promoting Centers Can Cause Intense Wakefulness

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- Discrete (individually separate) lesions in the *raphe nuclei* lead to a high state of wakefulness
- This is also true if bilateral lesions in the *medial rostral suprachiasmatic area in the anterior hypothalamus*
- In both instances, the excitatory reticular *nuclei of the mesencephalon* and *upper pons* seem to become released from inhibition, thus causing the intense wakefulness
- Indeed, sometimes lesions of the anterior hypothalamus can cause such intense wakefulness that the individual actually dies of exhaustion

Stages of Sleep

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Stages of Sleep

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Stage 1: is the transitional stage b/n drowsiness and sleep

- During this stage, muscles relaxed, respiration and pulse decreases
- This stage usually lasts only a few minutes, and if awakened, the person may say he or she was not asleep

Stage 2: is still a relative light sleep from which the person is easily wakened

- Rolling eye movements continue and snoring may occur

Stage 3 & 4: constitute “deep sleep”, sometimes termed as slow-wave sleep or delta sleep after characteristics waves seen on EEG

- During these stages the muscles are relaxed, but tone is maintained, respiration is even, and BP, pulse, and temperature decreases as do formation of urine and oxygen consumption of muscles
- Snoring, sleepwalking (somnambulism), and bed wetting (enuresis) are most likely to occur
- Strong stimuli are required to awaken people and dream contents tend to be realistic

- Polygraph recordings of electrophysiologic changes in brain waves (electroencephalogram; EEG),
- Eye movement (electrooculogram; EOG), and
- Muscle (electromyogram; EMG) show five sleep stages
- **Two main types of sleep are:**
 - ▣ NREM (non-rapid eye movement), quiet sleep and
 - ▣ REM (rapid eye movement) sleep

1- *Slow-wave sleep (NREM)*

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- In which the brain waves are strong and of low frequency
- This sleep is exceedingly restful and is associated with decreases in both peripheral vascular tone and many other vegetative functions of the body. For instance, there are 10 to 30 percent decreases in **BP, RR, and basal metabolic rate**
- Although slow-wave sleep is frequently called “**dreamless sleep**,” dreams and sometimes even nightmares do occur during slow-wave sleep

- The difference between the dreams that occur in slow-wave sleep and those that occur in REM sleep is that those of REM sleep are associated with more **bodily muscle activity**
- Also, the dreams of **slow-wave sleep** are usually **not remembered** because consolidation of the dreams in **memory does not occur**

2- REM Sleep (Paradoxical Sleep, Desynchronized Sleep)

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- **REM sleep:** closely resembles wakefulness except for very low muscle tone
- Respiration is irregular and oxygen consumption increases, vaginal secretion increases in females, and erection may occur in male
- Dreams occurring during REM sleep tend to be vivid and implausible, often including a sense of being unable to move

- In a normal night of sleep, bouts of REM sleep lasting 5 to 30 minutes usually appear on the average every 90 minutes
- When the person is extremely sleepy, each bout of REM sleep is short and may even be absent
- Conversely as the person becomes more rested through the night, the durations of the REM bouts increase

REM sleep has several important characteristics

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1. It is an active form of sleep usually associated with dreaming and active bodily muscle movements.
2. The person is even more difficult to arouse by sensory stimuli than during deep slow-wave sleep, and yet people usually awaken spontaneously in the morning during an episode of REM sleep
3. Muscle tone throughout the body is exceedingly depressed, indicating strong inhibition of the spinal muscle control areas.

4. Heart rate and respiratory rate usually become irregular, w/c is characteristic of the dream state
5. Despite the extreme inhibition of the peripheral muscles, irregular muscle movements do occur.
 - These are in addition to the rapid movements of the eyes
6. The brain is highly active in REM sleep, and overall brain metabolism may be increased as much as 20%

Cont...

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- The EEG shows a pattern of brain waves similar to those that occur during wakefulness
- This type of sleep is also called *paradoxical sleep* b/c it is a paradox that a person can still be asleep despite marked activity in the brain
- In summary, REM sleep is a type of sleep in which the brain is quite active
- However, the brain activity is not channeled in the proper direction for the person to be fully aware of his or her surroundings, and therefore the person is truly asleep

Sleep Rhythm

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- There is 90 minutes cycle during which people progress in sequence through the sleep stages
- The usual pattern is rapid progression through stages 1 to 4 and then back through stages 3 and 2, from which REM is then entered
- During the early part of the night, periods of slow wave sleep (stage 3 and 4) are longer
- In contrast, the time spent in rapid eye movement during the first cycle may be only 3 to 4 minutes;

- Whereas towards morning it may be as much as 45 minutes balanced with shorter periods of slow-wave sleep in which stage 4 may not be present
- If awakening occurs, the cycle begins again with stage 1, but if the awakening was brief; the tendency is to reenter the type of cycle from which the person was aroused
- Most people require 10 to 30 minutes to fall asleep; this period of time required is called ***sleep latency***

- A regular sleep latency of less than 5 minutes suggests excessive sleepiness;
- Sleep latency longer than 30 minutes may be accompanied by some sense of frustration with the time taken to get to sleep
- Short sleepers, that is those who sleep for less than 6 hours in 24Hrs, tend to be efficient, hardworking people;
- Long sleepers, that is those who sleep for more than 9 hours in 24Hrs, have a higher percentage of REM sleep and they are more creative

Altered Sleep/Rest Function

Potential for Altered Sleep

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- ❑ **Distraction**
- ❑ **Noise:** people chronically exposed to high noise levels have less slow-wave and REM sleep with more stage 1 and awakenings than their counterparts who live in quieter neighborhoods
- ❑ **Temperature:** excessive warmth or cold increases restlessness
- ❑ **Environment:** in new environment sleep latency is increased and total sleep time and proportion of REM are decreased

Altered Sleep/Rest Function...

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- ❑ **Illness:** during acute and chronic illness, clients are particularly vulnerable to loss of stage 3 sleep.
- ❑ The pain and discomfort of angina or dyspnea occurring during the night can disturb sleep.
- ❑ **Mood states:** *Anxiety* frequently delays sleep onset; *Depression* usually results in disturbed sleep
- ❑ Both the depression & sleep pattern disturbance may be linked to neurotransmitter imbalance

Altered Sleep/ Rest Function....

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- ❑ **Medication and chemicals:** sleep patterns are vulnerable to disturbance from medications taken to facilitate sleep, alcohol, medications used to treat other conditions, and other chemicals.
- ❑ Morphine, for example, increases the time spent awake during the sleep period and shortens total sleep time by decreasing both REM and stages 3 & 4

Altered Sleep/Rest Function....

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- ❑ **Lifestyle:** A fast-paced life filled with multiple stressors can result in the person's inability to relax easily or to fall asleep quickly
- ❑ **Diet:** Foods high in caffeine, such as coffee, colas, and chocolate, serve as stimulants and often disrupt the normal sleep cycle
- ❑ Also, consuming a large, heavy, or spicy meal just before bedtime may cause indigestion, which will likely interfere with sleep

Manifestation of Altered Sleep Function

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- **Signs and symptoms associated with sleep deprivation are:**
 - Fatigue,
 - Headache,
 - Nausea,
 - Increased sensitivity to pain,
 - Decreased neuromuscular coordination,
 - General irritability and
 - Inability to concentrate

- **Insomnia:** is a perceived difficulty in sleeping. Patterns of insomnia can be classified as onset insomnia (prolonged sleep latency), maintenance insomnia (multiple awakening) and early awakening insomnia
- **Narcolepsy:** is a disorder of excessive daytime sleepiness characterized by short, almost irresistible day time sleep attacks

- **Sleep apnea:** refers to recurrent periods of absence of breathing for ten seconds or longer, occurring at least 5 times per hour
- **Parasomnias:** are activities that are normal during activities but abnormal during sleep, such as sleep walking (somnambulism), talking, and bed wetting (enuresis)

Application of Nursing Process

Assessment

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- **Subjective data:** the single most important criterion for adequacy of sleep and rest is the client's statement
- **Functional pattern identification**
- *Determine the person's usual sleep and rest pattern through questions such as:-*
 - ▣ How many hours of sleep do you usually get?
 - ▣ What time do you usually go to bed? Get up?
 - ▣ What helps you get to sleep?
 - ▣ What makes it hard for you to sleep?
 - ▣ How do you feel when you awaken?

- ❑ **Risk identification**

- ❑ Assess caffeine, nicotine, and alcohol intake, and involvement in shift work.

- ❑ **Objective data**

- ❑ *Physical assessment*

- ❑ Observe for circles under the eyes, yawning, nodding, slowness of response, irritability, impaired concentration and word finding difficulties

Nursing Diagnosis

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- Sleep pattern disturbance
- *Defining characteristics*
 - ▣ Verbal complaint of difficulty falling asleep
 - ▣ Awakening earlier or later than desired
 - ▣ Interrupted sleep
 - ▣ Verbal complaints of not feeling well rested
 - ▣ Changes in behavior and performance – irritability, restlessness
 - ▣ Physical signs (slight hand tremor, frequent yawning)

Planning

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- **Specific client goals are:**
 - ▣ Client will report fewer problems with falling asleep
 - ▣ Client will report feeling more rested
 - ▣ Client will demonstrate physical signs of being rested

- *Nursing intervention to promote health and sleep/rest function*
- Environmental modifications, provision of intimacy and security, and sleep rituals are among the nursing interventions to promote health and sleep.
- *Nursing intervention for altered function*
- Rest: remains one of the most common symptomatic treatments for wide variety of sleep disorder.
- Hypnotics may be useful as a short term intervention during situational induced sleep pattern disturbance

Evaluation

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- The nurse evaluates the degree to which sleep pattern disturbance or inadequate rest has been resolved according to the client goals initially established
- *Goal:* Client will report fewer problems with falling asleep
- *Possible outcome criteria*
 - ▣ Within 7 days, client reports decrease in sleep latency to 10 to 15 minutes.
 - ▣ Within two days, client reports less anxiety regarding falling asleep.

THANK
YOU!!!!